

Roll No.

TCS-703

**B. TECH. (CSE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

COMPUTER NETWORKS—II

Time : Three Hours

Maximum Marks : 100

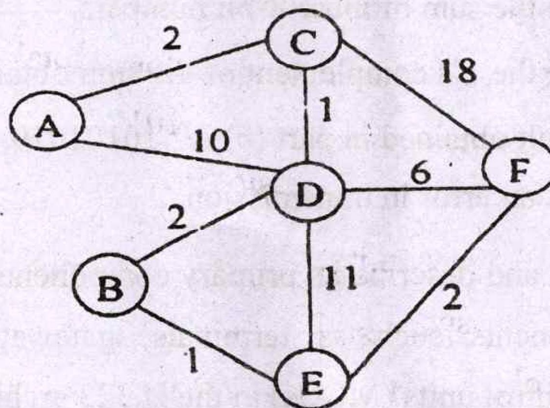
Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Consider the below network topology where circles represent routers and the routers are using link-state routing algorithm to determine the least cost routes to all other routers. Determine the routing table entries for routers B, C, D, E and F after convergence : (CO1, CO5)



P. T. O.

- (b) Discuss the advantages and potential limitations of the Link State algorithm in comparison to the Distance Vector algorithm. Provide examples of scenarios where the Link State algorithm might outperform the Distance Vector algorithm and vice versa. (CO1, CO5)
- (c) Consider the three node topology, the link cost are $c(x, y) = 6$, $c(y, z) = 7$, $c(z, x) = 4$, compute the distance tables after the initialization step and after each iteration of a synchronous version of the distance-vector algorithm. (CO1, CO5)
2. (a) Differentiate between the MAC (Media Access Control) address and the IP address. How are these addresses used in an Ethernet frame and why are they essential for network communication? (CO2, CO3)
- (b) Compare the effectiveness of CRC with other error detection techniques, such as checksums or parity checking. Highlight the advantages and limitations of CRC in ensuring data integrity. (CO2, CO3)
- (c) Given the following sequence of 8-bit binary numbers to be transmitted : (CO2, CO3)
- 11001100, 10101010, 11110000, 00111100
- (i) Calculate the sum of these 8-bit numbers.
- (ii) Calculate the 1's complement of the sum obtained in part (a).
- (iii) If the result obtained in part (b) is 10101010, determine whether there was an error in transmission.
3. (a) Define H.323 and describe its primary components. Explain the roles of these components such as terminals, gateways, gatekeepers, and multipoint control units (MCUs) in the H.323 architecture. (CO3, CO4)

(3)

- (b) Define BGP and elaborate on its primary function in internet routing. Explain the key differences between BGP and Interior Gateway Protocols (IGPs) like OSPF or RIP. (CO3, CO4)
- (c) Explain the practical applications of ARP and RARP in a networked environment. Provide scenarios where each protocol is used and the specific advantages they offer in those situations. (CO3, CO4)
- 4. (a) Elaborate RTSP and explain its role in multimedia streaming. Describe the primary functions of RTSP in streaming media delivery and control. (CO2, CO4)
- (b) Define SNMP and explain its primary purpose in network management. Describe the structure and elements of an SNMP-managed network, including SNMP managers, agents, and MIBs (Management Information Bases). (CO2, CO4)
- (c) Define the Session Initiation Protocol (SIP) and its significance in establishing, modifying, and terminating multimedia sessions. Explain the role of a SIP Registrar in SIP-based communication networks. (CO2, CO4)
- 5. (a) Explain the client-server interaction in a TCP Echo Server scenario. Detail the sequence of actions between a client and the server for data transmission and echoing. (CO5, CO6)
- (b) Write a C program for Passing Text Strings between Client and Server. (CO5, CO6)
- (c) What is Socket and what does it consist of ? Explain raise condition. (CO5, CO6)

Feb
18-12-23

Roll No.

TCS-704

**B. TECH. (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

ADVANCED COMPUTER ARCHITECTURE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Elaborate on Flynn's taxonomy. Provide examples of real-world systems that fall into each category, and explain the applicability of these categories in modern computing environments. (CO1)
- (b) The term "Instruction Set Architecture (ISA)" is often used when discussing computer architecture. Could you delve into the concept of ISA, its role in bridging the gap between hardware and software, and the significance of a well-defined ISA in computer system design ? (CO3)

P. T. O.

(c) State with brief description any *three* of the following : (CO1)

(i) Iron Law of Performance

(ii) Amdahl's Law

(iii) Lhadma's Law

(iv) Moore's Law

2. (a) Differentiate between read coherence and write coherence, the scenarios in which they are crucial, and the mechanisms used to maintain them.

(CO2)

(b) Design a simple cache memory system for a hypothetical processor. Assume a 32-bit address space, and a cache with 64 blocks, each block containing 16 bytes. The main memory has a total capacity of 512 MB. Explain how the cache is organized, including the cache index, tag, and offset. Illustrate how memory addresses are mapped to cache blocks, and how cache hits and misses are determined. Provide a step-by-step example of accessing memory addresses to demonstrate cache behavior.

(CO3)

(c) What is virtual memory, and how does it differ from physical memory (RAM) ? Could you explain the fundamental concept of virtual memory, its role in addressing memory limitations, and how it abstracts and extends physical memory ? (CO2), (CO4)

3. (a) What is a linear pipeline, and how does it differ from non-linear pipelines in terms of task execution and sequencing ? Can you provide an overview of the key characteristics of linear pipelines, including their fixed sequence of stages and the flow of tasks from one stage to the next ? (CO5), (CO6)

(b) How do branch hazards impact pipeline efficiency, and what are the strategies employed to mitigate these issues and maintain pipeline throughput? (CO2), (CO3)

(c) (i) List pipelining hazards with help of suitable example. (CO3)

(ii) Consider the following reservation table for a pipeline having three stages S₁, S₂ and S₃. Compute the minimal average latency (MAL)

Time →

	1	2	3	4	5
S ₁	×				×
S ₂		×		×	
S ₃			×		

4. (a) Write short notes on the following : (CO4)

- (i) Branch Prediction
- (ii) Direction Predictor
- (iii) Hierarchical Predictors

(b) What is Instruction-Level Parallelism (ILP) , and why is it important in optimizing program performance ? Can you provide an overview of the fundamental principles of ILP, including data dependencies and opportunities for parallel execution within a program ? (CO3), (CO5)

(c) Can you explain the different types of data hazards, including read-after-write (RAW), write-after-and write-after-write read (WAR), (WAW) hazards, and their causes ?

5. (a) In the context of parallel and distributed computing, what are the advantages and disadvantages of choosing message passing over shared memory or vice versa ? How do the characteristics of the application, system architecture, and scalability requirements influence the choice between these two paradigms ? (CO5)
- (b) Compare and contrast centralized and distributed shared memory architectures in the context of multiprocessor systems. Explain the concepts, advantages, and challenges associated with each approach. (CO3, CO5)
- (c) What is a vector architecture, and how does it differ from scalar architectures and traditional general-purpose processors ? Can you provide an overview of the key characteristics of vector processors. (CO5), (CO6)

Roll No.

TOE-707

B. TECH. (SEVENTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

CRYPTOGRAPHY AND NETWORK SECURITY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain security architecture. Justify its relevance and significance in providing security. (CO1)

(b) (i) Compare the monoalphabetic and polyalphabetic cipher. (CO1)

(ii) Alice wants to encrypt the message "Life is a journey not a destination" with transposition cipher. If Alice decides to use rail fence cipher with the key = 3, find the corresponding ciphertext for the message.

P. T. O.

- (c) (i) Eva have received a message "EXXEGOEXSRGI" from Bob.

Eva knows that Bob have encrypted the message using the Caesar cipher with key = 4, help Eva to retrieve the original message.

- (ii) Encrypt the message "My life is my message" with Caesar cipher using key value 3. (CO1)

2. (a) Compare substitution cipher with transposition cipher. Also explain why modern block ciphers are designed as combination of substitution ciphers and transposition cipher instead of transposition ciphers with the help of an example. (CO2)

- (b) (i) How many padding bits are added to a message of 100 characters if 8-bit ASCII is used for encoding and the block cipher accepts blocks of 64-bit. (CO2)

- (ii) Discuss Block cipher design principles in detail.

- (c) Analyze various methods of symmetric key distribution with respect to pros and cons involved with them. (CO3)

3. (a) How does public key cryptography differ from symmetric key cryptography. What are the requirements of public key cryptography. Also explain Diffie-hellman algorithm in detail. (CO3)

- (b) In an RSA cryptosystem, a particular user A uses two prime numbers $p = 3$ and $q = 11$ to generate his public and private keys. If the public key of A is $(7, n)$ and plaintext M is 31. Then find the private key of A and cipher text. (CO4)

(3)

- (c) Discuss the need of prime numbers in cryptography. Also find out whether 7 is a prime number or not, using Fermat's primality testing.
(CO4)
- 4. (a) Discuss the working principle of Kerberos V4 along with its limitations.
(CO5)
- (b) Discuss two IPsec protocols, ESP (Encapsulation Security Payload and AH (Authentication Header) in detail.
(CO5)
- (c) Compare one-way, two-way and three-way authentication schemes of X.509.
(CO5)
- 5. (a) Discuss SSL session and SSL connection in detail.
(CO6)
- (b) Categorize cloud security threats along with their countermeasures.
(CO6)
- (c) Discuss the need of the firewall along with its design principles. Also explain packet filter firewall with the help of a diagram.
(CO6)

(c) Discuss the need of prime numbers in cryptography. Also find out whether 7 is a prime number or not using Fermat's primality testing.

(CO4)

(a) Discuss the working principle of a cipher V along with its limitations.

(CO5)

(b) Discuss two types of protocols: ESP (Encapsulation Security Payload) and

(CO6)

AH (Authentication Header) in detail.

(c) Compare one-way, two-way and three-way authentication schemes of

(CO7)

X.509

(a) Discuss SSL session and SSL connection in detail.

(CO8)

(b) Categorize cloud security threats along with their countermeasures.

(CO9)

(c) Discuss the need of the firewall along with its design principles. Also

(CO10)

explain packet filter firewall with the help of a diagram.

Roll No.

TOE-702

**B. TECH. (CSE + CST+ CE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

SENSOR NETWORKS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Illustrate sensor network architecture in detail. List various modes of a sensor node. (CO1, CO2 & CO3)
- (b) What do you understand about sensors in WSN ? Explain types of sensor and its application in detail. (CO1, CO2 & CO3)
- (c) Outline the characteristic requirements and challenging issues in Wireless sensor networks and ad-hoc networks. (CO1, CO2 & CO3)
2. (a) Explain the classification of MAC protocol in detail. (CO2, CO3 & CO4)
- (b) Discuss and differentiate schedule-based protocol and random Access-based protocol in detail. (CO2, CO3 & CO4)
- (c) Write a note on the design goal of MAC protocol in wireless sensor networks. (CO2, CO3 & CO4)

P. T. O.

(2)

3. (a) Give a comparative analysis of hierarchical based routing and location base routing in sensor networks. (CO3, CO4 & CO5)
- (b) Wireless sensor network and wireless mesh network are part of the ad hoc wireless network. Justify that statement. (CO3, CO4 & CO5)
- (c) What is Data Centric routing ? "In Wireless Sensor Networks, data centric routing is used to control the redundancy of data." Justify this statement with a clear and concise explanation. (CO3, CO4 & CO5)
4. (a) Write short notes on the following : (CO3, CO5 & CO6)
 - (i) Current Trends in WSN
 - (ii) Machine Learning in WSN
- (b) Explain Operating system design issues for Wireless sensor networks. (CO3, CO5 & CO6)
- (c) Write a note on a flexible and application specific operating system namely TinyOS with respect to memory management, scheduling and resource sharing. (CO3, CO5 & CO6)
5. (a) What is the fundamental structure of a sensor node ? Provide instances that are appropriate to illustrate its usefulness in various fields. (CO3, CO5 & CO6)
- (b) Discuss in detail low-cost, low-data-rate wireless access technology namely IEEE 802.15.4. (CO3, CO5 & CO6)
- (c) Explain wireless sensor network applications and its impact on humans. (CO3, CO5 & CO6)

Roll No.

TOE-703

B. TECH. (CSE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023

INTELLECTUAL PROPERTY RIGHTS FOR ENGINEERS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is the difference between qualitative and quantitative research methods, and when is it appropriate to use each of them ? (CO1)
- (b) What is the significance of sampling in research, and what are the various sampling methods that researchers can use ? (CO2)
- (c) How can researchers ensure that their research findings are communicated effectively to a broader audience, including both academic and non-academic stakeholders ? (CO2)

P. T. O.

2. (a) What is the purpose of a literature review in a research study, and how can it be effectively conducted ? (CO1)
- (b) What are some common data collection methods used in research, and how do researchers choose the most appropriate method for their study ? (CO2)
- (c) What is the role of ethical considerations in research, and what are some key ethical principles that researchers must follow ? (CO3)
3. (a) What is the purpose of a technical report in the context of research, and how does it differ from a research paper ? (CO4)
- (b) What are the key components of a research proposal, and why are they important in the research process ? (CO2)
- (c) What is the role of a literature review in a research proposal, and how should it be used to establish the research's theoretical framework and rationale ? (CO2)
4. (a) What are intellectual property rights, and why are they important ? (CO5)
- (b) What are the main categories of intellectual property rights, and how do they differ ? (CO5)
- (c) What is the difference between a patent and a trademark, and when should one choose to apply for each ? (CO4)

(3)

5. (a) What are some common challenges and controversies related to intellectual property rights, particularly in the digital age ? (CO3)
- (b) What are the key steps for an individual or organization to enforce their intellectual property rights if they believe infringement has occurred ? (CO5)
- (c) How do intellectual property rights impact innovation and creativity, and what is the role of a patent system in fostering innovation ? (CO5)

(a) What are some common national and international treaties on intellectual property rights contained in the United States?

(b) What are the objectives of a national or international treaty on intellectual property rights contained in the United States?

(c) How do intellectual property rights impact innovation and development and what is the role of a patent system in fostering innovation? (100%)

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TOE-713

**B. TECH. (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

ARTIFICIAL INTELLIGENCE AND PATTERN RECOGNITION

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss various Artificial Intelligence techniques in detail.

(CO1, CO2 & PO1)

(b) What are the various types of Reasoning in Artificial Intelligence.

(CO1 & PO2)

(c) What is an Intelligent Agent in Artificial Intelligence ? Discuss its various types with examples.

(CO1, CO2 & PO1)

2. (a) What do you mean by Logic Programming ? Write a Logic Program to check whether a given number is prime or not.

(CO2, CO5 & PO2)

(b) Write a short note on Bayesian Networks in Artificial Intelligence.

(CO1 & PO4)

(c) What are Rule Based Systems in AI ? Give an example of a Rule Based System.

(CO2 & PO3)

P. T. O.

(2)

3. (a) Explain the concept of Pattern Recognition in AI with a suitable example. (CO3 & PO1)
- (b) What are the different types of Pattern Recognition Algorithms in AI. Explain each in short. (CO1, CO2 & PO2)
- (c) Discuss the K-Nearest Neighbour Classification Algorithm in detail. (CO2 & PO1)
4. (a) What do you understand about Unsupervised Machine Learning ? Give some examples of unsupervised machine learning algorithms in short. (CO1 & PO2)
- (b) Discuss the application of pattern recognition in the field of Medicine. (CO4 & PO1)
- (c) Explain the Fuzzy Classification algorithm in Pattern Recognition. (CO4, CO1 & PO2)
5. (a) What are the various types of Neural Networks ? Explain each in short. (CO1, CO3 & PO4)
- (b) Explain the Architecture of Artificial Neural Networks in detail. (CO1, CO2 & PO1)
- (c) Explain the Backpropagation Learning Algorithm with a suitable example. (CO6 & PO2)

Roll No.

TCS-706/TOE-714

B. TECH. (CSE) (SEVENTH SEMESTER) END SEMESTER EXAMINATION, Dec., 2023

ARTIFICIAL INTELLIGENCE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is an expert system ? What are the main advantages in keeping the knowledge base separate from the control module in knowledge-based systems ? (CO4)
- (b) Explain the concept of utility-based agent with an example. (CO4)
- (c) Define the terms 'Fact and 'Rule'. Express 'A car without wheels is not valuable' in predicate logic. (CO4)
2. (a) What are the differences and similarities between problem solving and planning ? (CO3)

P. T. O.

(2)

- (b) How is forward chaining different from backward chaining inference method ? (CO3)
- (c) Distinguish between predicate and propositional logic with suitable example. (CO3)
- 3. (a) Explain A* searching technique in detail with an example. Discuss the conditions for optimality of this technique. (CO1)
- (b) Discuss the steps involved in natural language processing with an example. (CO1)
- (c) What are parametric and non-parametric model ? What do you understand by the hyperparameter ? (CO1)
- 4. (a) What do you mean by PEAS ? Determine the PEAS values for the following AI problems : (CO2)
 - (i) Vacuum Cleaner
 - (ii) Autonomous taxi
- (b) What is min-max algorithm for game playing ? (CO2)
- (c) What is parsing ? Parse the sentence "Kathy jumped the horse" using top-down parsing. (CO2)
- 5. (a) Explain AI agents design philosophy using neat and clean diagrams. (CO1)
- (b) Discuss different techniques of pattern recognition using suitable example. (CO1)
- (c) What is Speech Recognition ? Explain with the help of block diagram. (CO1)

Roll No.

TCS-750

B. TECH. (CSE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023
CLOUD ORCHESTRATION AND LOAD BALANCING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Waverton Investment Management boosts business resiliency and sustainability with Microsoft Azure. Evaluate AWS Cloud automation services for Managing VMware Cloud from perspective of above mentioned organization. (CO1)
- (b) Bank Credit Dnipro secures banking services for customers with Microsoft Azure amidst instability. Evaluate Google Cloud Load Balancing from perspective of above mentioned organization. (CO1)

P. T. O.

- (c) Vietnamese bank builds the country's first cloud-native mobile banking app on Azure. Evaluate Cloud automation and orchestration difference from perspective of above mentioned organization. (CO1)
2. (a) WirelessCar virtually removed cold starts and improved the performance of its connected mobility services at no extra cost by embracing AWS Lambda SnapStart for Java. Analyze Use cases of AWS CloudFormation, and IBM Cloud Orchestrator from perspective of above mentioned organization. (CO2)
- (b) Isabel Group automates operations, takes efficiency to the next level with Azure data platform. Analyze Automated Security and Compliance from the perspective of above mentioned organization. (CO2)
- (c) WorkApps transformed its broadcast messaging solution, follow. us, using Amazon SQS, AWS Media Services, and Amazon Translate, ensuring seamless integration, high scalability, and the ability to broadcast millions of messages in second. Analyze Terraform Architecture with working from the perspective of above-mentioned organization. (CO2)
3. (a) Polish fin-tech company mElements develops country's first cloud-native payment gateway with Azure. Analyze AWS Autoscaling types and benefits from the perspective of above mentioned organization. (CO3)

- (b) Leading funds manager in the Philippines ATRAM achieves faster data processing with Azure Synapse Analytics. Analyze Ingram Micro Cloud Orchestrator and IT Automation with AWS Lambda from the perspective of above-mentioned organization. (CO3)
 - (c) Dutch Financial Services Company, Rabobank, Creates Global Data Platform with Azure Synapse Analytics and Power BI. Analyze AWS Systems Manager Automation Limitations and OpenStack Heat orchestration engine from the perspective of above-mentioned organization. (CO3)
4. (a) Using fully managed AWS services, financial technology company ekonoo SA achieved regulatory approval for its pension management solution and can focus on delivering value to its customers. Analyze Heat orchestration service from the perspective of above-mentioned organization. (CO4)
- (b) Luxury fashion company Tapestry built a scalable IaC platform to seamlessly deploy modernized workloads in a nimble, consistent, and repeatable manner for security and governance, helping the company reduce deployment times, maintain security and compliance protocols, and drive innovation. Analyze MapReduce and HDFS from the perspective of above-mentioned organization. (CO4)
 - (c) Fintech lending platform Aro increased its efficiency using managed search clusters on Amazon OpenSearch Service, creating more capacity to innovate. Analyze the Apache Storm from the perspective of above-mentioned organization. (CO4)

5. (a) Travis Perkins built on Amazon Web Services (AWS) as part of its strategic effort to create an agile and innovative business and to free data siloed in legacy infrastructure for better insights. Analyze the Representational State Transfer and It's API from the perspective of above-mentioned organization. (CO5 & CO6)
- (b) Moody's Senior Vice President of Cloud Engineering, Divya Elaty, discusses how Moody's leverage AWS services like AWS Glue, Amazon Dynamo DB, Amazon S3, and AWS Step Functions to build modern data solutions and custom in-house tools to meet the company's data needs. Analyze RESTful API authentication methods from the perspective of above-mentioned organization. (CO5 & CO6)
- (c) FOX has improved its marketing and advertising insights and ML capabilities by building a unified data solution using Amazon S3, Amazon Rekognition, and other cloud technology. Analyze the steps for Smart Traffic Management System Application to Cloud using RESTful API from the perspective of above mentioned organization. (CO5 & CO6)

Roll No.

TCS-771/TCS-742

B. TECH. (CSE) (SEVENTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

NATURAL LANGUAGE PROCESSING WITH BIG DATA

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain any *four* ways of converting text to features in NLP. (CO1)
(b) Illustrate vector space model, and cosine similarity measure in natural language processing. (CO1)
(c) With an example, show how an n-gram model can improve a natural language processing task. (CO2)
2. (a) Using Python, write suitable code to create IF and TF-IDF weights of three sentences stored in string variables s1, s2, s3. (CO2)
(b) Briefly analyze any *two* linear classifiers used in NLP. How does a classifier's fit() method differ from predict(). (CO3)

P. T. O.

(2)

- (c) Explain the terms pretrained language models and model fine tuning in natural language processing. (CO5)
3. (a) Differentiate between contextual embeddings and static embeddings with the help of an example of each type. (CO3)
- (b) Explain language modelling and next sentence prediction tasks in BERT model pretraining. (CO5)
- (c) How are recurrent neural networks (RNNs) used as language models ? Explain with the help of any *one* RNN. (CO4)
4. (a) How is Naïve Bayes classifier used to classify Spam vs. Ham messages ? What is the need of Laplace smoothing in NB classifier ? (CO2)
- (b) Out of 100 samples, 60 are positive and 40 are negative. A classifier is used and it classifies 50 correct positive samples and 35 correct negative samples. Create the confusion matrix and calculate precision, recall and F1-score. (CO2)
- (c) Illustrate the terms cross-validation, and stratified sampling with the help of suitable examples. (CO2)
5. (a) Discuss any *four* natural language processing applications where deep learning is used. (CO6)
- (b) Explain loss function, gradient, and backward differentiation used in training neural networks. (CO3)
- (c) With an example, explain how self-attention mechanism is used in natural language processing. (CO5)

Roll No.

TCS-731

**B. TECH. (CSE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

COMPUTER FORENSICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Provide a comprehensive definition of cybercrime and offer a real-world example of a cybercrime incident. Explore the impact of cybercrimes on individuals, organizations, and society. (CO1)
- (b) Distinguish between malware, viruses, and worms, and discuss recent global cybersecurity threats. (CO1)
- (c) Delve into the topics of cyberterrorism and information warfare in detail. (CO1)

P. T. O.

2. (a) Define digital forensics and examine the opportunities and challenges within this field. Offer a critical analysis. (CO2)
- (b) Create a theoretical chain of custody for a digital evidence case. (CO2)
- (c) Explain the forensic data acquisition process and clarify the concept of 'order of volatility' in data acquisition. (CO2)
3. (a) Elaborate on the steps involved in establishing a 'Chain of Custody' for evidence, emphasizing the importance of meticulous documentation and its role in preserving evidence integrity during investigations. (CO3)
- (b) Discuss the reasons for creating duplicates of acquired memory images and outline best practices for data acquisition. (CO3)
- (c) Describe the concept of watermarking and its application in hardware security. (CO3)
4. (a) What are the traditional problems associated with finding digital evidence ? (CO4)
- (b) Analyze data hiding techniques used in the concealment of digital evidence, discussing steganography, encryption, and other methods. Evaluate their relevance and potential challenges for digital forensics professionals. (CO4)
- (c) Develop a comprehensive case study scenario involving a cybercrime, integrating multiple aspects of digital forensics, and outline the steps a forensic investigator should follow to solve the case. (CO4)

(3)

5. (a) Investigate a hypothetical case in which an individual host obscene profiles online. Analyze the legal implications, ethical considerations, and technical challenges associated with this case, and propose an investigative plan. (CO5)
- (b) Discuss the complexities and nuances of conducting live forensics in the context of a cybercrime investigation. Explore the challenges, tools, and best practices for extracting digital evidence from live systems while preserving system integrity. (CO5)
- (c) Conduct an extensive analysis of a case involving credit card fraud, providing a detailed account of the digital evidence required to investigate and prosecute the perpetrators. Discuss the steps involved in handling the case from a digital forensics perspective. (CO5)

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20/12/23
7th sem
9:30 to 12:30

Roll No.

TCS-761

**B. TECH. (CSE) + ALL/B. TECH. (BIOTECH)
(SEVENTH SEMESTER)**

END SEMESTER EXAMINATION, Dec., 2023

CLOUD INFRASTRUCTURE SERVICES

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss the various issues and challenges associated with Cloud Infrastructure. (CO1)
- (b) Describe the NIST cloud computing architecture. (CO1)
- (c) Explain the need of Cloud management. Specify the activities involved in cloud management. (CO1)
2. (a) Discuss the Cloud Infrastructure Management Interface (CIMI) model proposed by DMTF. How CIMI is implemented in your cloud ? (CO3)

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(2)

- (b) Explain the steps involved in cloud computing life cycle management implementation. (CO3)
- (c) Discuss the issues and challenges to be addressed for implementing Infrastructure Security at the Host Level in Cloud Computing. (CO3)
- 3. (a) What is SOA ? Discuss the various components of SOA. (CO3)
- (b) What is service catalogue ? How SOA services are monitored and managed ? (CO3)
- (c) Describe the relationship between Cloud Computing and Service Oriented Architecture (SOA). (CO3)
- 4. (a) Explain the concept of Cloud storage in the Digital Universe. (CO2)
- (b) Explain the Cloud Data Management Interface (CDMI) proposed by SNIA. (CO2)
- (c) Discuss the various types of cloud storage and interoperability feature. (CO2)
- 5. (a) Discuss any case study related to Amazon ElastiCache. Justify its usability. (CO4)
- (b) Explain the concept and working of Amazon Route 53. (CO4)
- (c) Discuss any case study related to usage of Amazon redshift. (CO4)

Roll No.

TIT-721

B. TECH. (CSE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023

BUSINESS INTELLIGENCE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss about an Industry of your choice such as retail, manufacturing, telecommunications, finance etc. Identify few common core business functions. (CO1 & CO2)
- (b) What is an enterprise IT application ? Describe it through typical architecture with suitable diagram. (CO1 & CO2)
- (c) What are the characteristics of Internet-ready IT Applications with its several advanced common capabilities. (CO1 & CO2)
2. (a) Give an example of data being collected and stored in an unstructured format in your University. (CO1 & CO2)

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(2)

- (b) Can XML data be converted into a structured format ? Explain with an example. (CO1 & CO2)
- (c) Discuss the techniques to manage Unstructured data ? How to extract information from stored Unstructured data ? (CO1 & CO2)
- 3. (a) State about the system (OLTP/OLAP) has support for concurrency control and recovery and why. (CO1 & CO3)
- (b) Write some queries that cannot process by OLTP system. Should OLAP be performed directly on operational database ? Explain (CO1 & CO3)
- (c) Construct an ER Diagram to demonstrate the working of a bank. Make all the necessary assumptions. (CO1 & CO3)
- 4. (a) Explain these terms in William H. Inmons definition of a data warehouse : Subject oriented, integrated, non-volatile and time variant. (CO4 & CO6)
- (b) Assume you are a project manager who has been sent to collect business requirements for a retail chain. Give a few examples of business requirement that you would have collected. (CO4 & CO6)
- (c) Discuss about the two logical design techniques for data modelling. What could be the problem posed by the ER modelling. (CO4 & CO6)
- 5. (a) Why is it important for enterprises to go for “enterprise dashboard”. Explain. (CO5 & CO6)
- (b) Explain the ETL process. Mention few ETL tools available in the market. (CO5 & CO6)
- (c) What are the benefits of integrating social media information into BI applications ? What are its drawbacks ? (CO5 & CO6)

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TCS-756/TOE-701

B. TECH. (CSE/CE/CST/AI & DS) (SEVENTH SEMESTER) END SEMESTER EXAMINATION, Dec., 2023

HUMAN COMPUTER INTERACTION

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What characteristics make an interface successful ? List the human elements that need to be considered while designing an HCI and include a suitable diagram. (CO1 & CO2)
- (b) Describe the contributions that design, psychology, computer science, and engineering have made to the growth and development of HCI. (CO2)
- (c) Comparing an AI-enabled user interface with a suitable example against a traditional user interface that does not use AI. (CO2)

P. T. O.

2. (a) One example is the vertical scroll bar found in word processing programs and web browsers. Provide a single instance of how the scrollbar complies with each of Norman's five design principles. Give a brief explanation of the interface model for the scroll bar. (CO2)
- (b) Give an example of a normal user interaction with the site's presented information structure. In this activity, there are two distinct Cognitive Dimensions that stand out. You should discuss each one and how it affects the user's experience. (CO4)
- (c) Analyze, using an example, how people's lives at work are affected by their interactions with computers. (CO5)
3. (a) Could you explain the steps involved in the UI Design and Development Process ? Give an example using a helpful figure. (CO4)
- (b) What are the design and development goals of HCI ? Describe these tools in detail. CO1
- (c) Instructions and appointment reminders should have the least complicated visual design possible to ensure safety when riding. Using formal components of visual design, list three specific ways this can be accomplished. (CO5)
4. (a) Explain in detail about the following Drag and Drop elements : (CO3 & CO4)
 - (i) Drag and Drop Module
 - (ii) Drag and Drop List
 - (iii) Drag and Drop Object
 - (iv) Drag and Drop Action
 - (v) Drag and Drop Collection

(3)

- (b) Provide an example of how operable controls work and discuss the benefits and drawbacks of using buttons. (CO2 & CO3)
 - (c) How can I select colors in an HCI ? Define the problem of color usage as well. (CO3)
5. (a) Discuss the Keyboard and function keys, pointing devices, speech recognition digitization and generation with the help of suitable live example. (CO5)
- (b) Create a Case Study on How to Improve Programming or Researcher Tools Using HCI Software Tool Methods. (CO6)
 - (c) Examine and classify user models, user assistance, sock-organizational problems, and stakeholder needs for HCI systems. (CO6)

(a) Provide an example of how standard controls work and discuss the

benefits and drawbacks of such controls.

(b) How can I detect controls in an HCI? Define the problem of color usage

as well.

(c) Discuss the keyboard and function keys pointing devices, speech

recognition, digitization and gesture with the help of suitable live

examples.

(d) Create a Case Study on how to improve performance in HCI.

Topic: Human-Computer Interaction.

(e) Examine and discuss user models, task analysis, and organizational

problems and risk factors for HCI.

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TCS-785

**B. TECH. (CSE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

DEEP LEARNING FOR COMPUTER VISION

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are some common techniques used for feature extraction in image processing ? Provide an example of an application of feature extraction in computer vision. (CO1 & CO2)
- (b) How can histogram equalization be used to improve the performance of histogram equalization in situations where the images are noisy ? Explain with a suitable example. (CO1 & CO2)
- (c) How Arithmetic/Logic Operations are performed in Digital image processing ? Explain with a suitable example for each Arithmetic/Logic Operations. (CO1 & CO2)

P. T. O.

2. (a) Write a Pseudo code for image subtraction in the wavelet domain.
(CO2 & CO3)
- (b) How image segmentation is utilized to identify the objects in computer vision applications ? If the images are blurry, what will be the impact on the results of segmented images ? Discuss with a suitable example.
(CO2 & CO3)
- (c) Discuss the process of image-fusion using deep learning in medical images with a suitable framework.
(CO2 & CO3)
3. (a) Explain the concept of a neural network and its basic components in the context of deep learning. Also describe the purpose and function of activation functions in a neural network.
(CO3 & CO4)
- (b) What is the primary purpose of using convolutional layers in CNNs for computer vision tasks ? Explain the terms “stride” and “padding” in the context of convolutional layers. Provide an explanation and demonstrate the potential differences by considering 7×7 image matrix.
(CO3 & CO4)
- (c) Why second order derivative are utilized to enhance the image quality ? Find first and second order derivative for following data : (CO3 & CO4)
- | | | | | | | | | | |
|----|----|---|---|---|---|---|----|----|---|
| 32 | 32 | 0 | 5 | 1 | 5 | 1 | 17 | 17 | 5 |
|----|----|---|---|---|---|---|----|----|---|
4. (a) Explain how motion detection is incorporated into the background subtraction technique. How does motion detection assist in differentiating moving objects from the static background in videos ?
(CO4 & CO5)

(3)

- (b) Design and discuss a framework using deep learning approach to develop automatic activity recognition for any real time application.

(CO4 & CO5)

- (c) Perform the Huffman code for below image matrix to obtain the average code length :

(CO4 & CO5)

1	1	0	4	0
0	0	0	1	2
1	2	1	1	3
4	0	2	0	3
0	4	0	1	0

5. (a) Consider a 7×7 gray scale image with pixel intensities ranging from 0 to 255, apply k -means clustering with $k = 2$. Initialize the cluster centers as [100, 200], and provide the final cluster centroids after convergence.

(CO5 & CO6)

- (b) Describe the architecture of a typical Convolutional Neural Network (CNN). Discuss the advantages and challenges of using pre-trained models for new computer vision tasks. Justify your answer with a suitable example.

(CO5 & CO6)

- (c) Consider a binary image of a circle in a 6×6 grid, with the circle's center at (4, 4). Apply dilation using a 3×3 circular structuring element. How does the dilated image appear ? Also discuss how the original image was altered by each operation.

(CO5 & CO6)

